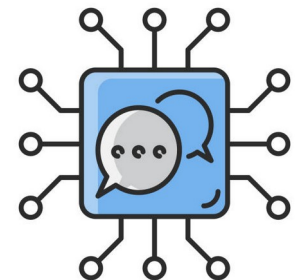


Anatomy of Language

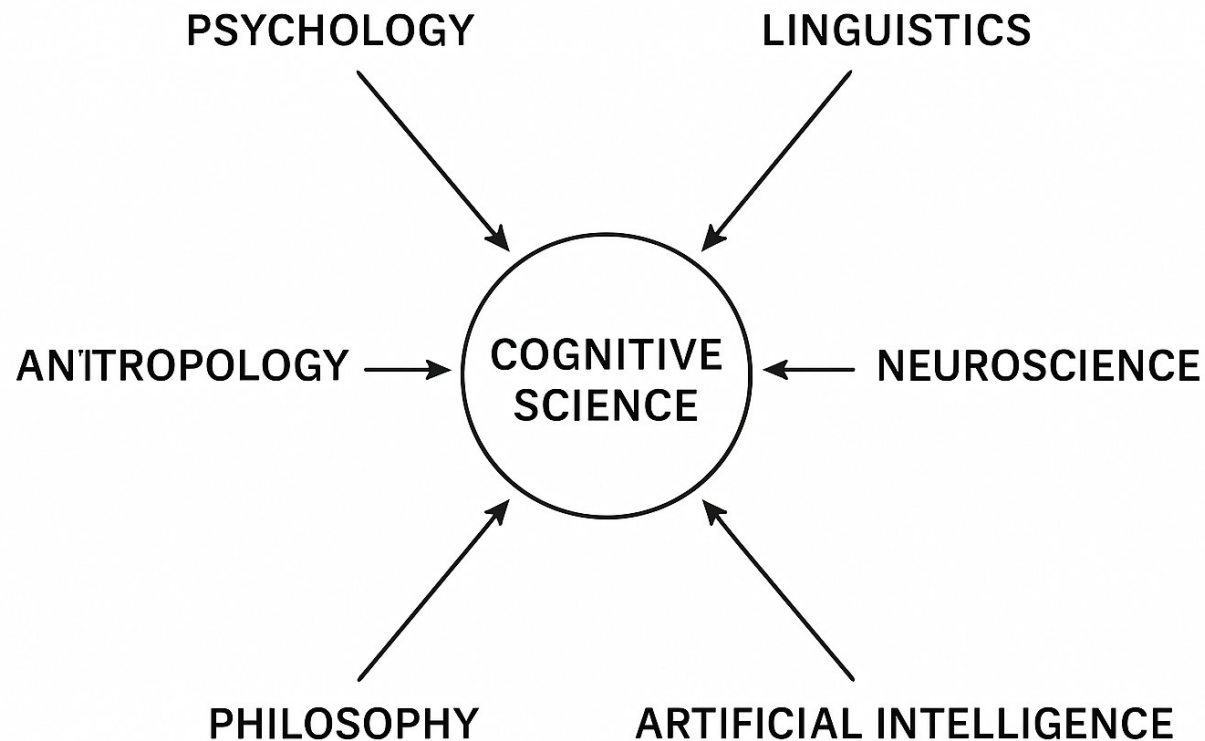
Tushar B. Kute,
<http://tusharkute.com>



Anatomy of a Language

- What is Language? (Definitions)
- Language as a Rule-Governed, Dynamic System
- The Innateness of Grammar
- The Three Facets: Biological, Social, Psychological
- Modes of Language: Spoken vs. Written
- Expression (Signifier) & Content (Signified)
- Natural vs. Artificial Languages
- Linguistics as a Scientific Study

Cognitive Science



Language



What is Language?

- Is it just communication? (No, animals communicate).
- Is it just words? (No, grammar is key).
- A simple definition: "A structured system of communication used by humans, based on grammar and a lexicon."
- A cognitive definition: "A computational system of the mind that allows us to build and understand complex structures for thought and expression."

Language

- Language is a structured system of communication. The structure of a language is its **grammar** and the free components are its **vocabulary**.
- Languages are the primary means of communication of humans, and can be conveyed through spoken, sign, or written language.
- Many languages, including the most widely-spoken ones, have writing systems that enable sounds or signs to be recorded for later reactivation.
- Human language is not dependent on a single mode of transmission (sight, sound, etc.) and is highly variable between cultures and across time.

What Makes Human Language Unique?

- Duality of Patterning: A limited set of sounds (phonemes) creates a massive set of words.
- Arbitrariness: The word "dog" has no inherent connection to the animal (it's not "woof").
- Productivity (Creativity): We can create and understand sentences never uttered before. (e.g., "The green alien is eating a blue taco on Mars.").
- Displacement: We can talk about the past, the future, and things that don't exist (e.g., "yesterday," "next week," "a unicorn").

What Makes Human Language Unique?

Duality of Patterning

A limited set of sounds (phonemes) creates a massive set of words



Arbitrariness



The word „dog” has no inherent connection to animal (it's not „woof")

Productivity (Creativity)

[A] + [B] → [C]

We can create and understand sentences never uttered before (e.g. "The green alien is eating a blue taco on Mars.")

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We can talk about the past, the future, and things that don't exist (e.g. "yesterday", "next week," a unicorn')

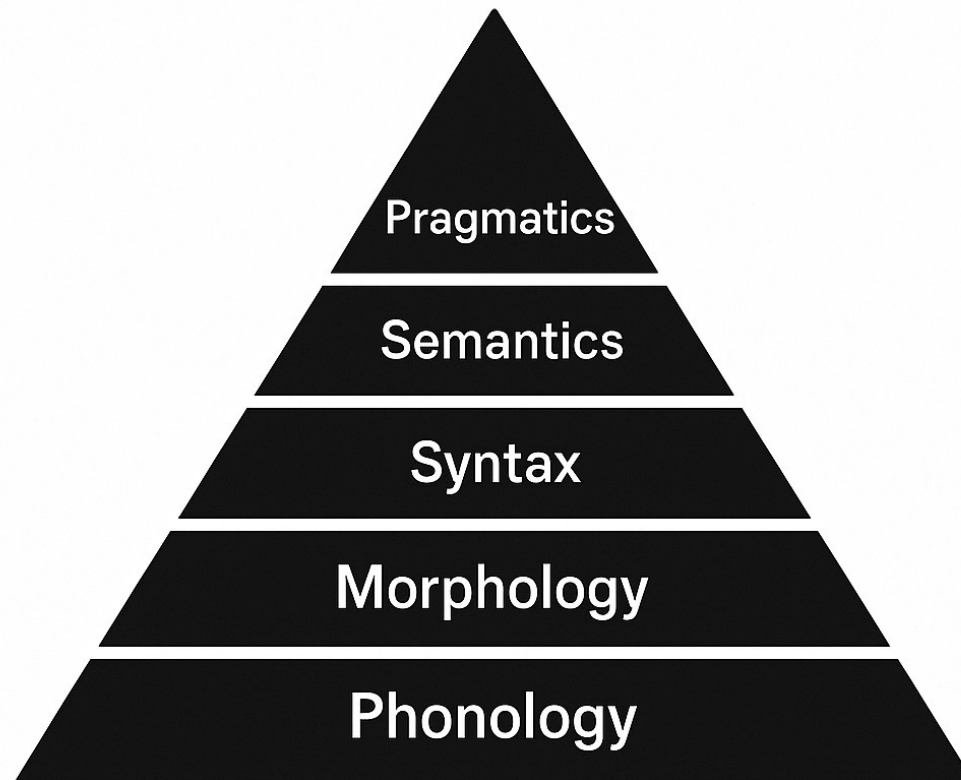
Section 2:

Language as a Rule-Governed, Dynamic System

Language as a Rule-Governed System

- Language is not random; it's governed by a complex set of rules (our grammar).
- These rules are mostly implicit and unconscious. We know them without knowing we know them.
- Example: You know "The big red car" is correct, and "The red big car" is... weird. (This is a syntactic rule of adjective order).
- This is why NLP is hard: we must make these implicit rules explicit for a computer.

The "Rulebook": Levels of Linguistic Analysis



The "Rulebook": Levels of Linguistic Analysis

- Phonology: Rules of sound systems (/p/ vs. /b/).
- Morphology: Rules of word formation (walk + ed = walked).
- Syntax: Rules of sentence structure (Subject-Verb-Object).
- Semantics: Rules of meaning (What does "bank" mean? River bank? Money bank?).
- Pragmatics: Rules of context (Saying "Can you pass the salt?" is a command, not a question).

Language as a Dynamic System

- Language is not a fixed, static object. It is a living, evolving system.
- It changes across:
 - Time (Diachronic Change): Old English -> Middle English -> Modern English.
 - Space (Dialects): UK English ("lift," "boot") vs. US English ("elevator," "trunk").
 - Social Groups (Sociolects): Slang, jargon, formal vs. informal.
- New words are born ("google" as a verb, "selfie"), and old words die.

How Language Evolves

- Lexical Change:
 - "Nice" used to mean "silly" or "ignorant" (in the 1300s).
 - "Awful" used to mean "full of awe" or "awe-inspiring."
- Syntactic Change:
 - Word order used to be more flexible (Old English).
 - Loss of "thou" / "thee" for the simpler "you."
- This dynamic nature is a major challenge for NLP models, which must be constantly retrained.

Section 3:

Knowledge of Language: Innateness of Grammar

How Do We Know Language?

- A child learns their native language fluently by age 5. How?
- The "Nurture" Camp (Empiricism):
 - B.F. Skinner (Behaviorism).
 - Language is learned entirely through imitation, reinforcement, and correction.
 - The mind is a "blank slate" (tabula rasa).
- The "Nature" Camp (Nativism):
 - Noam Chomsky (Linguist).
 - Language is an innate biological property of humans.
 - We are born with a "language blueprint."

Chomsky's Revolution: Universal Grammar (UG)

- Chomsky argued that Behaviorism fails to explain language acquisition.
- He proposed a Language Acquisition Device (LAD): a hypothetical module of the brain, present at birth, that allows infants to acquire language.
- This LAD contains the Universal Grammar: a set of core principles and parameters common to ALL human languages.
- (e.g., All languages have nouns and verbs; all languages have ways to ask questions).

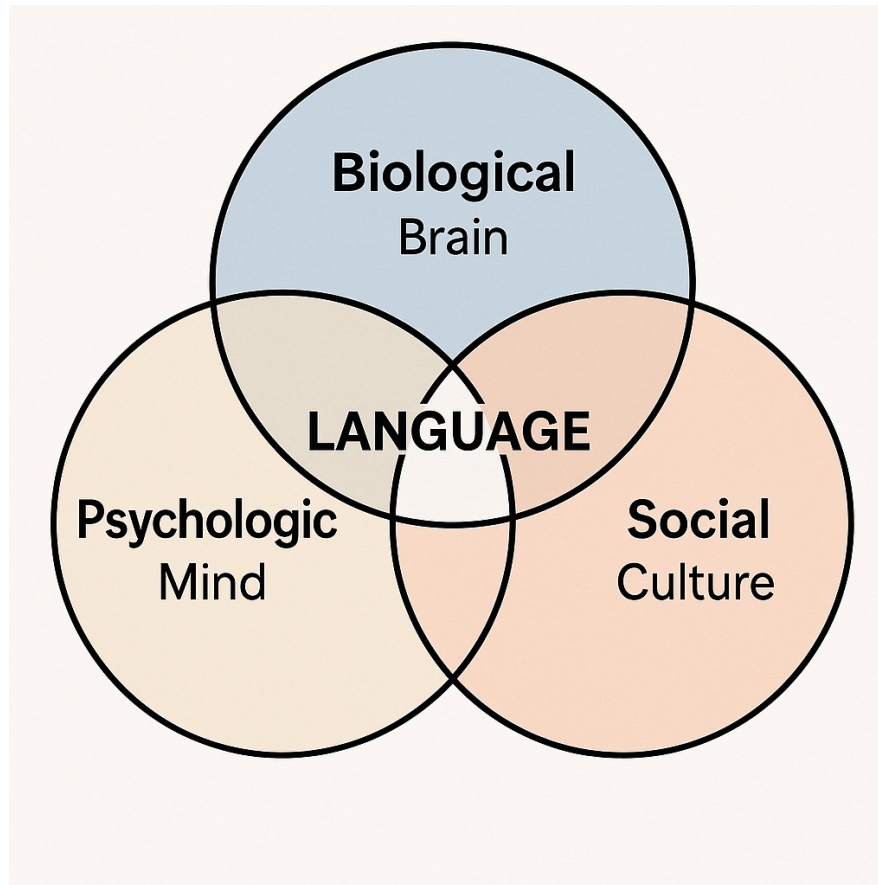
Evidence: The Poverty of the Stimulus

- This is Chomsky's core argument.
- Children hear a finite, messy, and incomplete set of language data (the "stimulus").
- ...Yet they produce an infinite, systematic, and complete grammatical system (their "knowledge").
- Conclusion: The knowledge must be richer than the input. The gap is filled by our innate Universal Grammar.
- Example: Children inherently know the complex rule for forming questions, even for complex sentences, without ever being taught it.

Evidence: Critical Period & Creolization

- The Critical Period Hypothesis:
 - There is a limited window in childhood (birth to puberty) when language acquisition is effortless.
 - After this period, it becomes extremely difficult (as seen in "feral children" cases like Genie). This suggests a biological, not purely learned, process.
- Creolization:
 - When adults from different language backgrounds are mixed, they form a "Pidgin" (a simple, broken code).
 - Their children, however, spontaneously and uniformly develop this Pidgin into a full-blown, grammatically rich language called a "Creole." They invent grammar.

A Tri-Partite Phenomenon



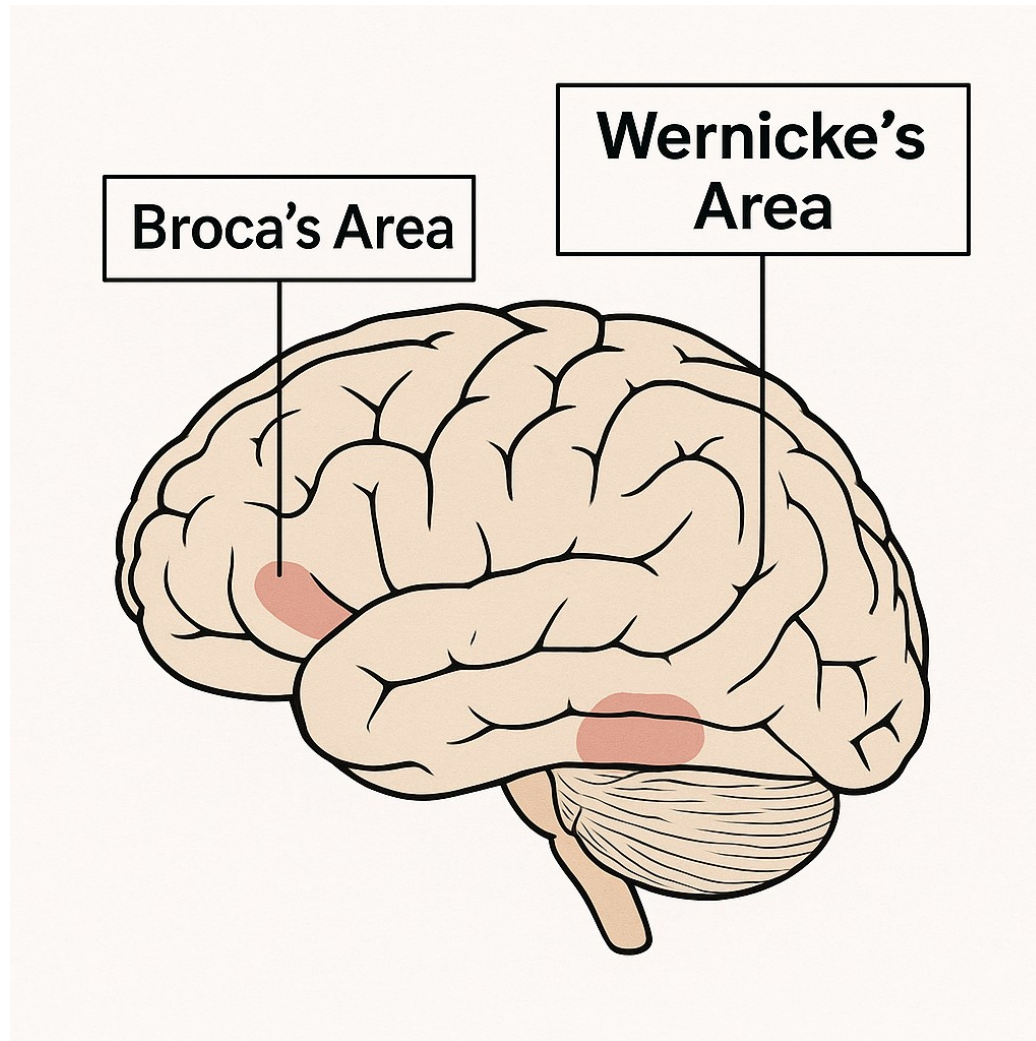
A Tri-Partite Phenomenon

- To understand language, we must see it as three things at once:
 - Biological: A product of our brains ("the hardware").
 - Psychological: A product of our minds ("the software").
 - Social: A product of our societies ("the network").

The "Hardware": Language & The Brain

- Language is not just "anywhere" in the brain. It's highly localized.
- Broca's Area: (Frontal lobe) Responsible for speech production and syntactic processing. Damage here leads to Broca's Aphasia: difficulty speaking, but (mostly) intact comprehension.
- Wernicke's Area: (Temporal lobe) Responsible for language comprehension. Damage here leads to Wernicke's Aphasia: fluent, grammatical speech that is semantically meaningless ("word salad").

The "Hardware": Language & The Brain



The "Software": Language & The Mind

- This is the domain of Psycholinguistics.
- How do we process language in real-time?
- Lexical Access: How do you hear a sound and instantly retrieve the word "cat" from a mental dictionary of ~50,000 words?
- Parsing: How does your mind build a "syntax tree" as you read this sentence, assembling meaning on the fly?
- This is what NLP tries to replicate computationally.

The "Network": Language & Society



- This is the domain of Sociolinguistics.
- Language is a social contract. We agree that "dog" means dog.
- It's our primary tool for social interaction: building relationships, signaling identity, and transmitting culture.
- Your language use (accent, word choice) signals who you are, where you're from, and which groups you belong to.

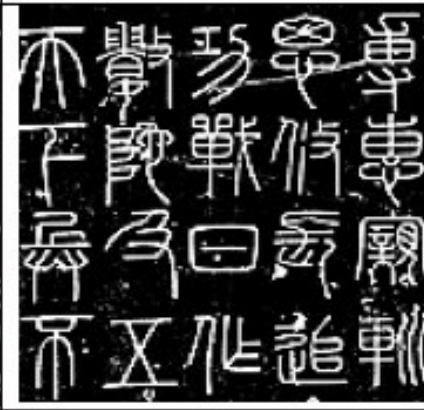
Section 5:

Modes of Language (Spoken vs. Written)

Modes of Language: Spoken vs. Written

- Language manifests in different modes or "channels."
- Primacy of Speech:
 - Spoken language is the primary, biological form.
 - Humanity had speech for 100,000+ years.
 - Written language is a ~5,000-year-old technology.
 - Every human culture has speech; not all have writing.
 - We acquire speech naturally; we must be taught writing.

Written Language

Egypt Hieroglyph	Chinese Character	Indus Script	Sumer Cuneiform
			

Spoken vs. Written: A Comparison

- Spoken Language:
 - Transient (disappears as it's said)
 - Contextual (shared time/place)
 - Has Prosody (tone, pitch, stress)
 - Spontaneous, interactive, less formal
- Written Language:
 - Permanent (an artifact)
 - Decontextualized (reader is absent)
 - Uses Punctuation
 - Planned, non-interactive, more formal

Spoken vs. Written: A Comparison

Spoken Language	Written Language
• Transient (disappears as it's said) • Contextual (shared time/place) • Has Prosody (tone, pitch, stress) • Spontaneous, interactive, less formal	• Permanent (an artifact) • Decontextualized (reader is absent) • Uses Punctuation • Planned, non-interactive, more formal

A Note on Sign Language

- Sign Language (e.g., ASL, ISL) is NOT a "code for" spoken language.
- It is a full, natural language in its own right.
- It is visual-gestural, not auditory-vocal.
- It has its own complex grammar (phonology, morphology, syntax) and exhibits all the same properties (Productivity, Arbitrariness, Duality).
- This confirms that the "language faculty" in the brain is abstract and not tied to any single mode.

Section 6:

Expression and Content (Semiotics)

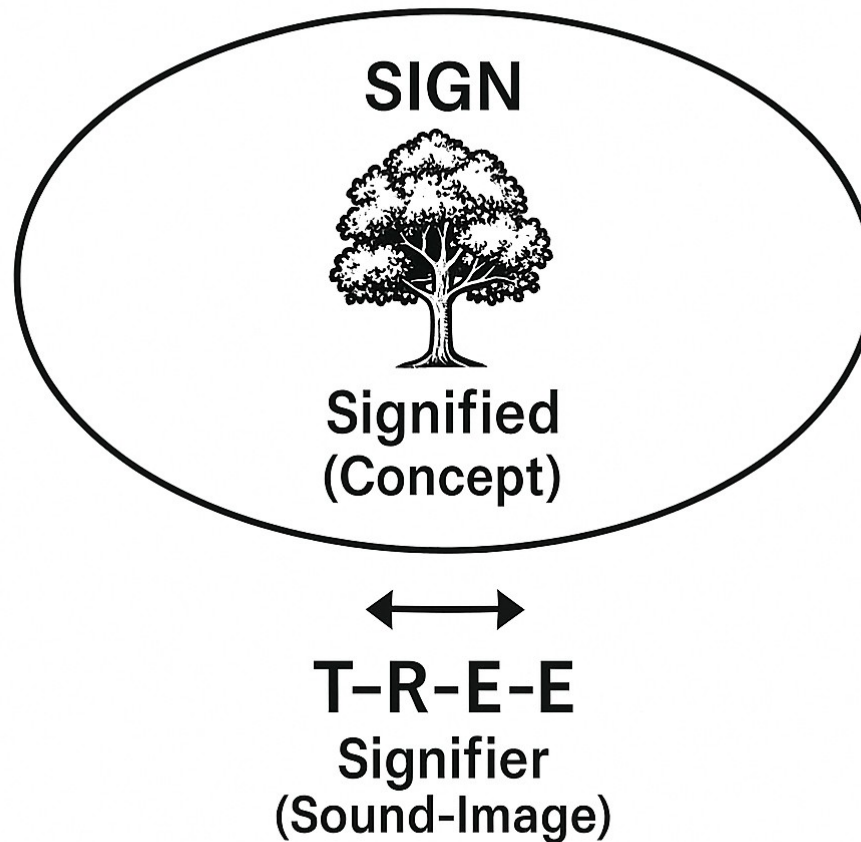
Language as Expression & Content

- This is the foundation of Semiotics (the study of signs).
- Ferdinand de Saussure, a Swiss linguist, proposed that a Sign (like a word) is a two-part entity.
- It's NOT just a label for a "thing."
- It's a link between a concept and a sound-image.

Expression (Signifier) & Content (Signified)

- Signified (Content): The concept or idea in your mind. (e.g., the idea of a four-legged, barking animal).
- Signifier (Expression): The sound-image or form used to represent that concept. (e.g., the sound "d-o-g").
- The link between them is arbitrary (as seen in Slide 5).
- The "Sign" is the whole unit (concept + form).

Expression (Signifier) & Content (Signified)



Expression vs. Content

- Louis Hjelmslev (Danish linguist) expanded this.
- He said both Expression and Content have their own Form and Substance.
- Example: The sound /p/
 - Expression Substance: The actual physical puff of air, the lip movement.
 - Expression Form: The abstract rule in English that /p/ is different from /b/.
- NLP Connection:
 - Early NLP focused on Expression Form (syntax, rules).
 - Modern NLP (Word2Vec, BERT) tries to model Content Form (the "concept space" or semantic relationships).

Section 7:

Natural vs. Artificial Language

Language as a Symbolic System

- Language is the ultimate symbolic system.
- A symbol is something that stands for something else (e.g., + stands for addition; "cat" stands for the concept of a cat).
- There are two major classes of symbolic language systems:
 - Natural Languages (NL)
 - Artificial Languages (AL)

Natural Language

- Languages that have evolved naturally in human societies.
- Examples: English, Marathi, Mandarin, Swahili, ASL.
- Key Properties:
 - Evolved, not designed.
 - Ambiguous: "I saw a man on a hill with a telescope."
(Who has the telescope?)
 - Redundant: "The dog, he..." (Useful for processing).
 - Socially embedded and acquired effortlessly by children.

Artificial Language

- Languages that are designed by humans for a specific, formal purpose.
- Examples:
 - Logical Languages: Predicate Calculus, Boolean Algebra.
 - Programming Languages: Python, Java, C++.
 - Constructed Languages: Esperanto, Klingon.
- Key Properties:
 - Designed, not evolved.
 - Unambiguous: if $(x > 5)$ can only mean one thing.
 - Efficient & Precise: No redundancy.
 - Formally defined and must be explicitly learned.

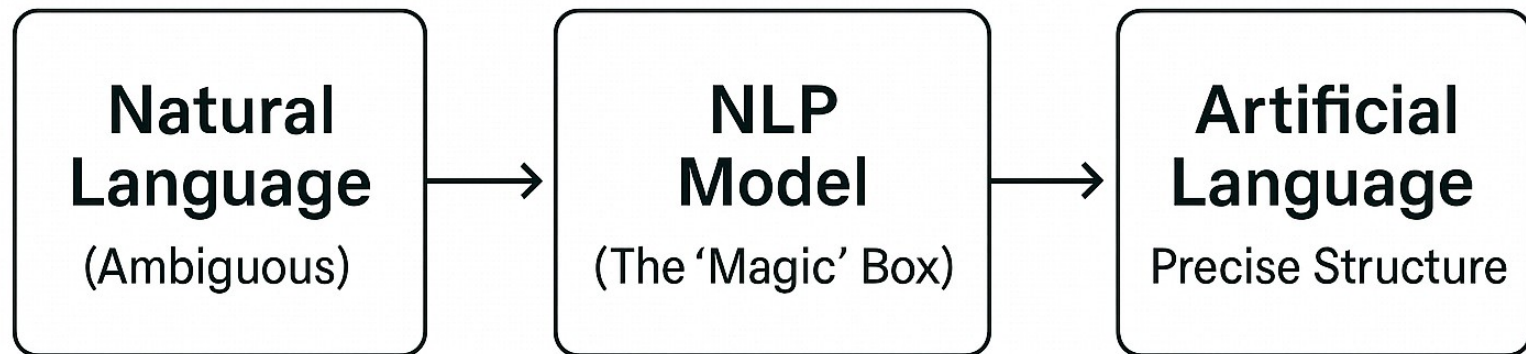
Comparison: NL vs. AL

Feature	Natural Language (e.g., English)	Artificial Language (e.g., Python)
Origin	Evolved over millennia	Designed for a purpose
Ambiguity	Highly Ambiguous (a feature!)	Unambiguous (by design)
Rules	Implicit, complex, flexible	Explicit, rigid, formal
Acquisition	Acquired by children	Learned by students
Purpose	Social, expressive, thought	Formal, computation, logic
Redundancy	High (e.g., "that man, <i>he</i> left")	Low (e.g., <code>print()</code>)

The Challenge of NLP

- NLP is the process of teaching computers to understand and process Natural Language (NL).
- Computers only understand Artificial Language (AL).
- The entire field of NLP is a "translation service": How do we map the ambiguous, redundant, and contextual rules of NL onto the unambiguous, efficient, and formal rules of AL?

The Challenge of NLP



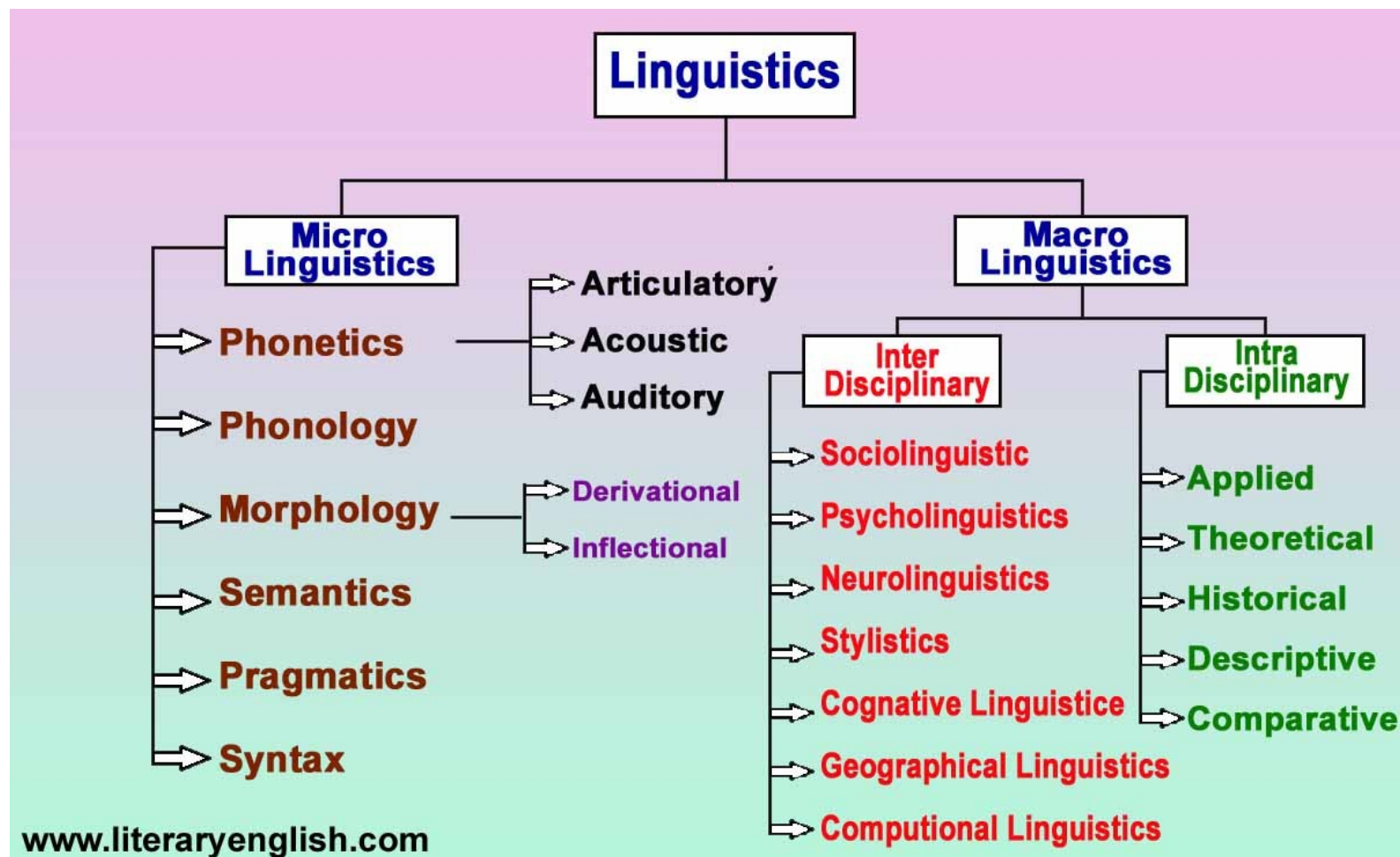
Linguistics as a Scientific Study

- Linguistics is NOT the art of learning many languages (that's polyglotism).
- Linguistics is the science of language.
- It seeks to answer:
 - "What is language?"
 - "What are the universal properties of language?"
 - "How does the human mind store and process it?"
 - "How and why does it change?"

How Do Linguists "Do Science"?

- Observation: Listen to speakers, analyze texts (corpora), find patterns.
 - Observation: "I walk," "You walk," but "He walks."
- Hypothesis: Form a rule.
 - Hypothesis: "In English, 3rd person singular verbs in the present tense get an '-s'."
- Testing: Try to break the hypothesis (find counterexamples).
 - Test: "He can walk." (Fails!)
- Refine Theory: The rule is not for all verbs, only main verbs, not modal auxiliary verbs.
- This descriptive, evidence-based approach makes it a science, just like biology or chemistry.

Linguistics



www.literaryenglish.com

Linguistics

- Linguistics looks at:
 - The general phenomenon of human language.
 - Different families of languages (example: Germanic, including English, German, Dutch and Scandinavian, among others).
 - Specific languages (example: Arabic, Mandarin and French).
 - Communicative codes or behaviors that are not so well defined (example: the language of recent immigrants, the ways by which bilinguals choose one or another language in certain settings).

The "Lab" of Linguistics

- Core ("Theory"):
 - Phonology, Morphology, Syntax, Semantics
- Interdisciplinary ("Applied"):
 - Cognitive Linguistics: Language & the mind.
 - Sociolinguistics: Language & society.
 - Neurolinguistics: Language & the brain.
 - Historical Linguistics: Language change.
 - Computational Linguistics (NLP): Language & computers.

Why This Matters for You (NLP-CV)

- You cannot build a system to process language if you don't respect what language is.
- "All models are wrong, but some are useful."
- Knowing linguistics helps you understand why your NLP model is failing.
 - Is it a syntax problem? (Parsing)
 - Is it a semantics problem? (WSD)
 - Is it a pragmatics problem? (Context)
- Linguistics provides the problem definition that Machine Learning tries to solve.

Session 1: Key Takeaways

- 1. Definition: Language is a productive, arbitrary, and displaced symbolic system.
- 2. Rules: It's a rule-governed system (phonology, syntax, etc.) but also dynamic (it evolves).
- 3. Innateness: Humans are born with a Universal Grammar (Chomsky), which explains the "poverty of the stimulus."

Session 1: Key Takeaways

- 4. Facets: It's simultaneously biological (brain), psychological (mind), and social (culture).
- 5. Modes: Speech is primary; writing is a secondary technology.
- 6. Semiotics: Words are "signs" linking Content (Signified) to an Expression (Signifier).
- 7. NL vs. AL: NLP's job is to translate ambiguous Natural Language into formal Artificial Language.
- 8. Science: Linguistics is the scientific study of these phenomena.

Next Session...

- We will move from the "what" to the "how."
- Topics:
 - Paradigmatic & Syntagmatic Relationships
 - Form, Function, and Meaning
 - Deep dive into the Levels of Linguistic Analysis

References:

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Thank you

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